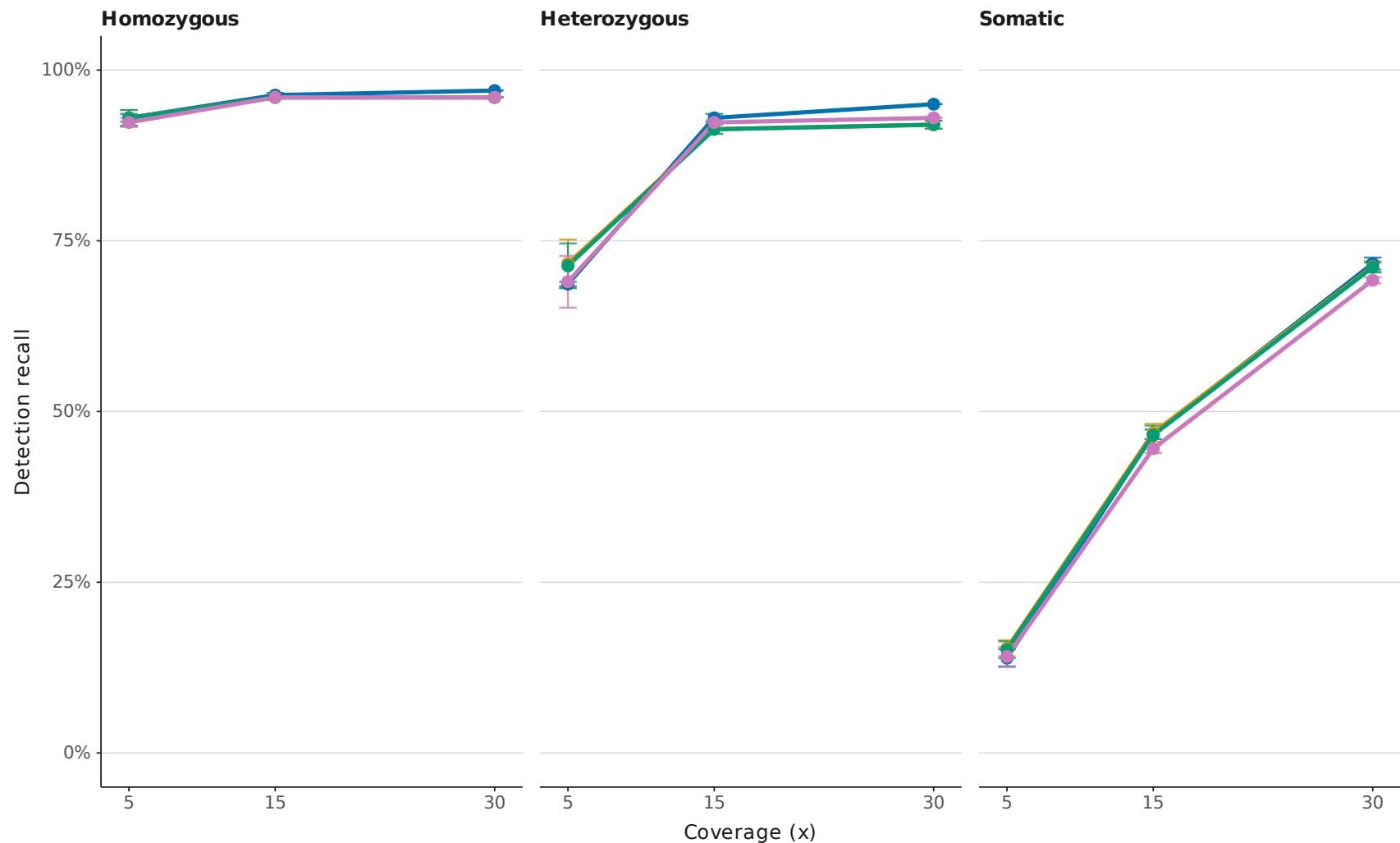


Detection recall vs. coverage

Fraction of simulated insertions recovered (mean $\pm$ SEM, 3 replicates)

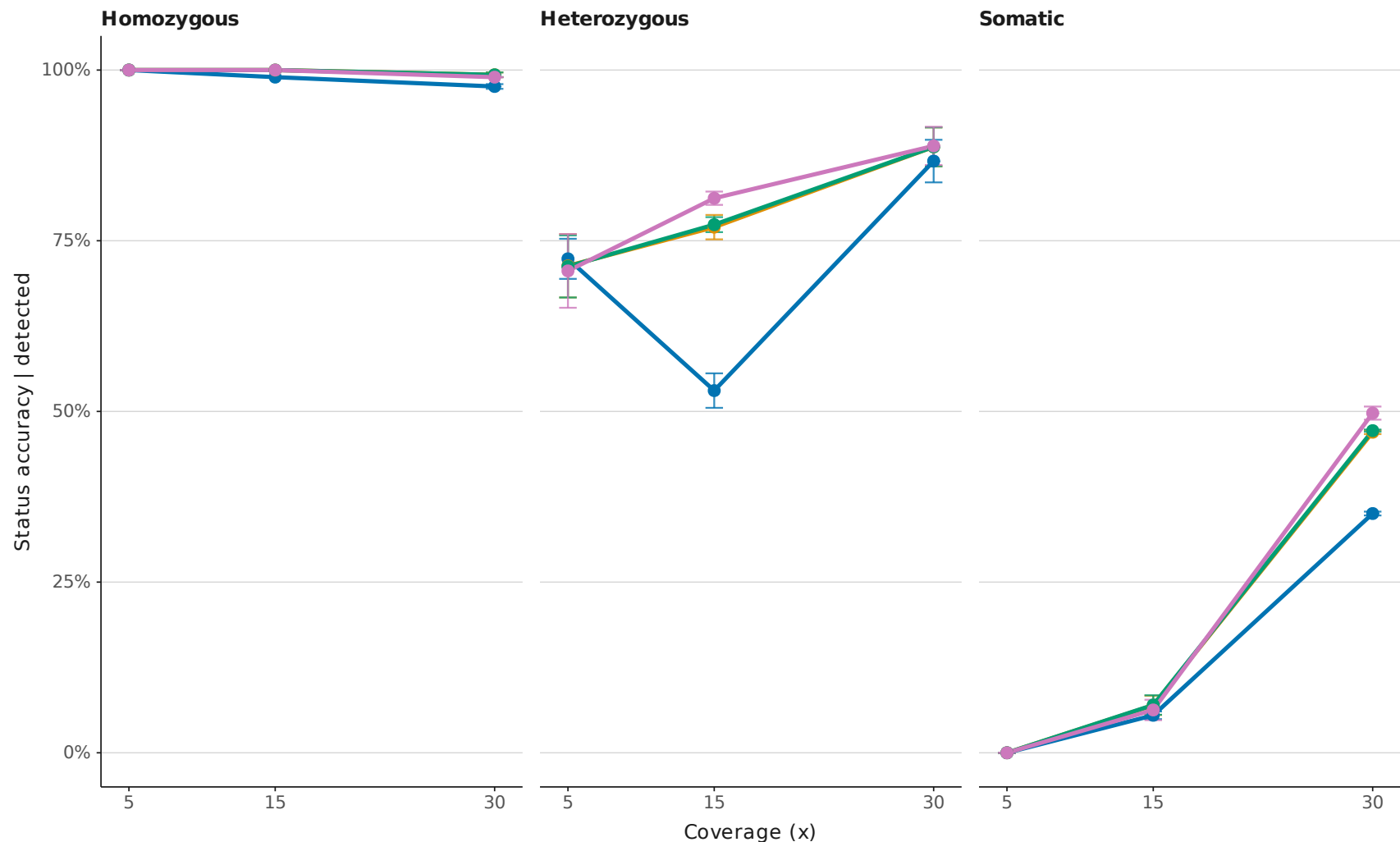
RelocaTE2 RelocaTE3-bowtie2/bwa RelocaTE3-bwa/bwa RelocaTE3-minimap2/minimap2



Genotype-status accuracy vs. coverage

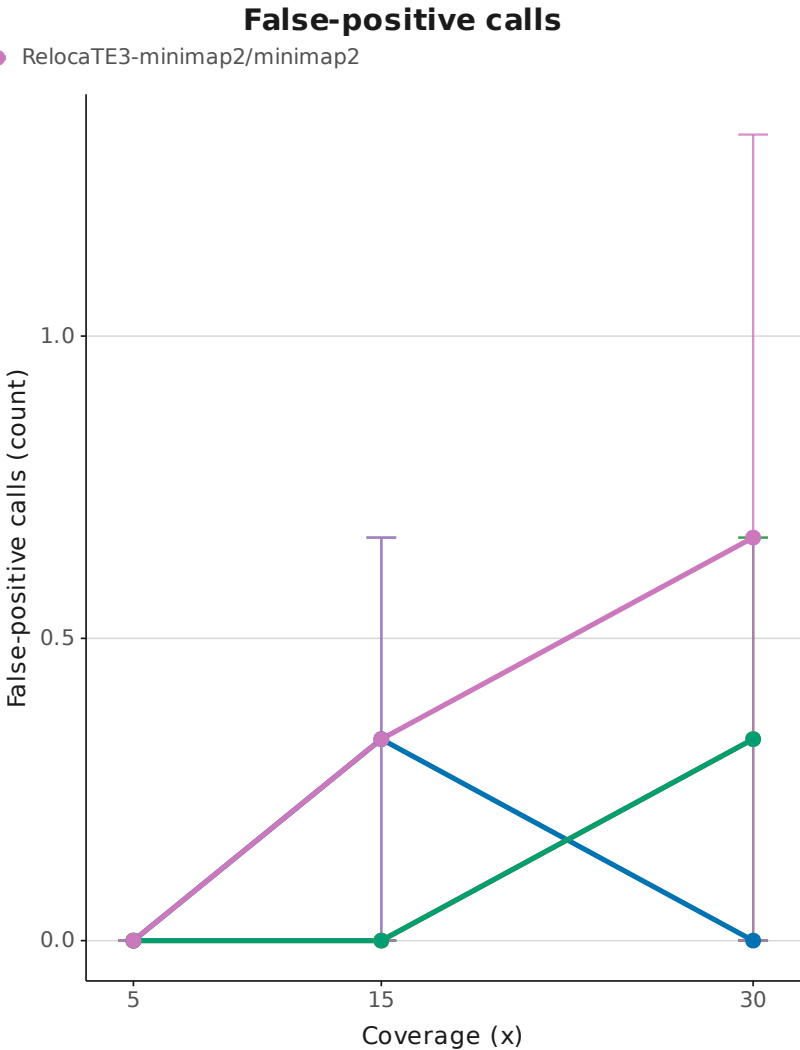
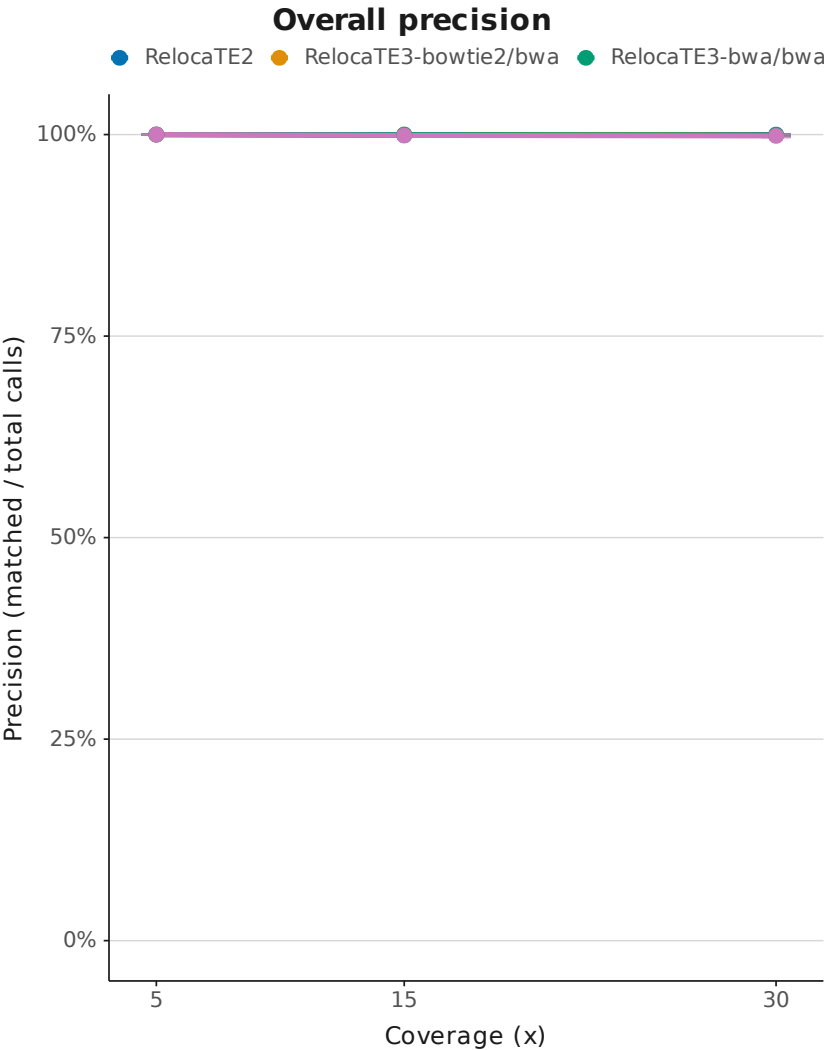
Correct hom/het/somatic call among DETECTED insertions (mean $\pm$ SEM)

RelocaTE2 RelocaTE3-bowtie2/bwa RelocaTE3-bwa/bwa RelocaTE3-minimap2/minimap2



Precision and false positives vs. coverage

RelocaTE precision = fraction of calls matching a true insertion (mean $\pm$ SEM)



Somatic detection recall by cellular fraction

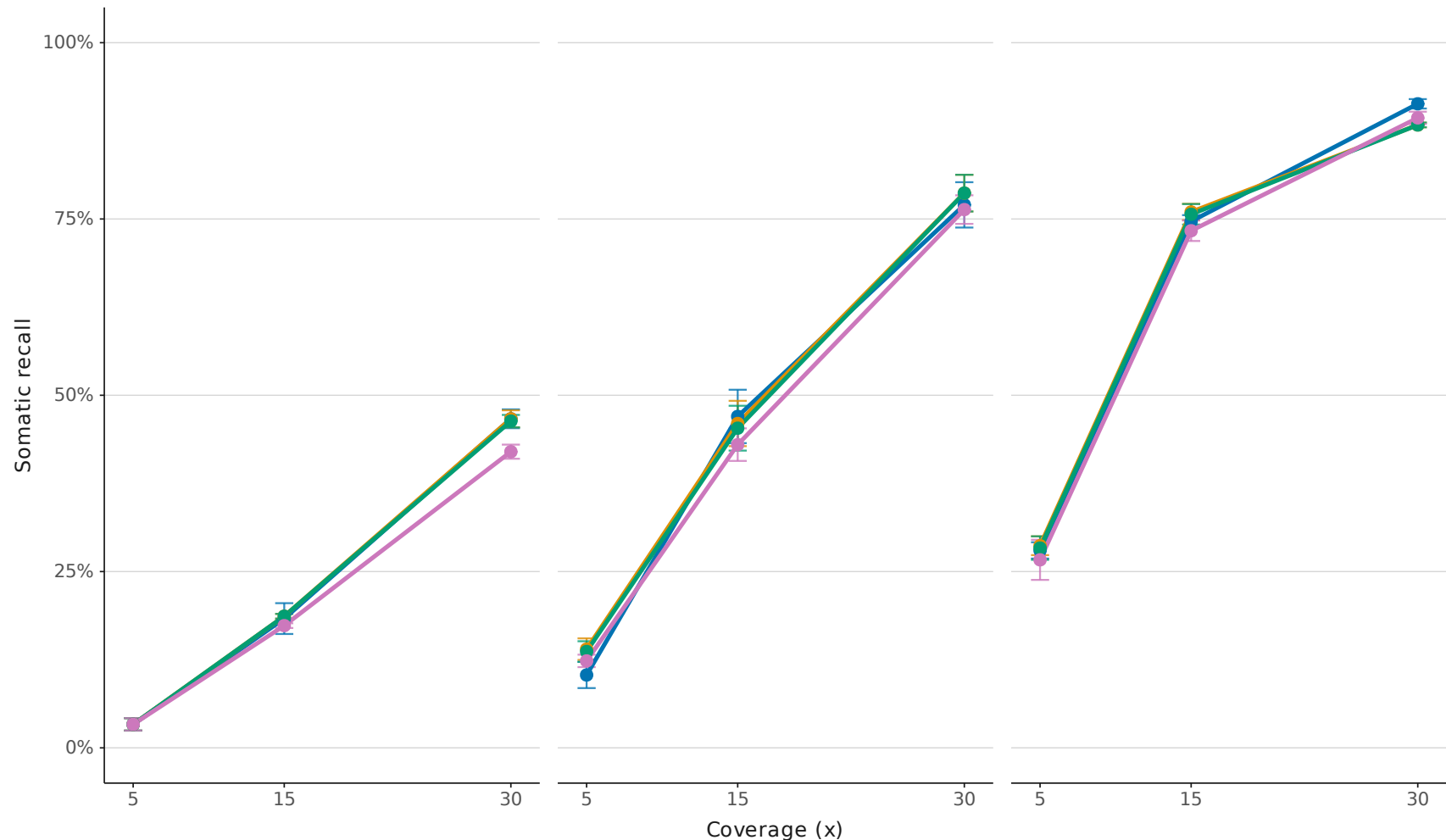
Lower cellular fraction = lower variant allele frequency = harder (mean $\pm$ SEM)

● RelocaTE2 ● RelocaTE3-bowtie2/bwa ● RelocaTE3-bwa/bwa ● RelocaTE3-minimap2/minimap2

Cellular fraction 0.1 (VAF 0.05)

Cellular fraction 0.2 (VAF 0.1)

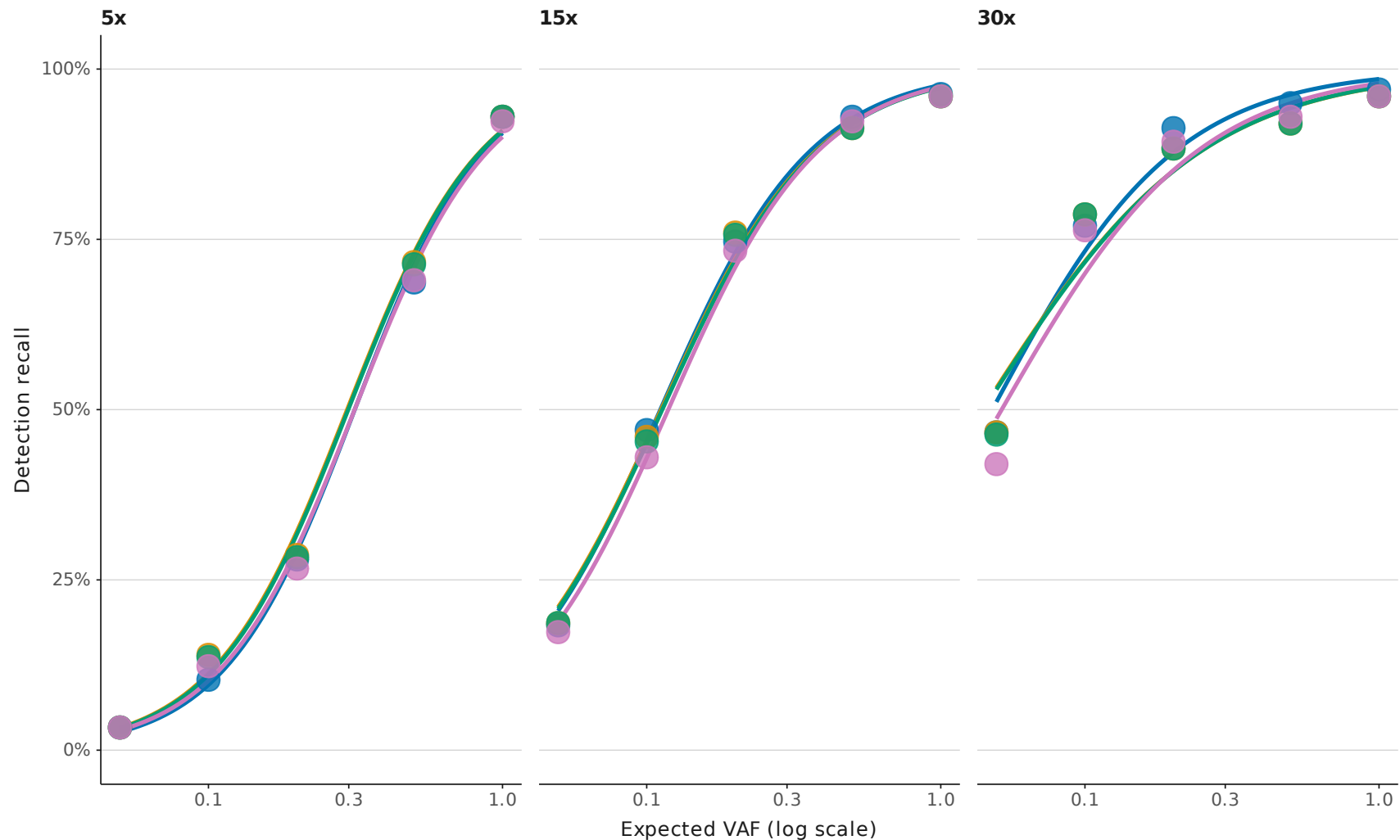
Cellular fraction 0.4 (VAF 0.2)



Detection sensitivity vs variant allele frequency

Logistic fit per caller; VAF at 50% detection = limit of detection (LOD50)

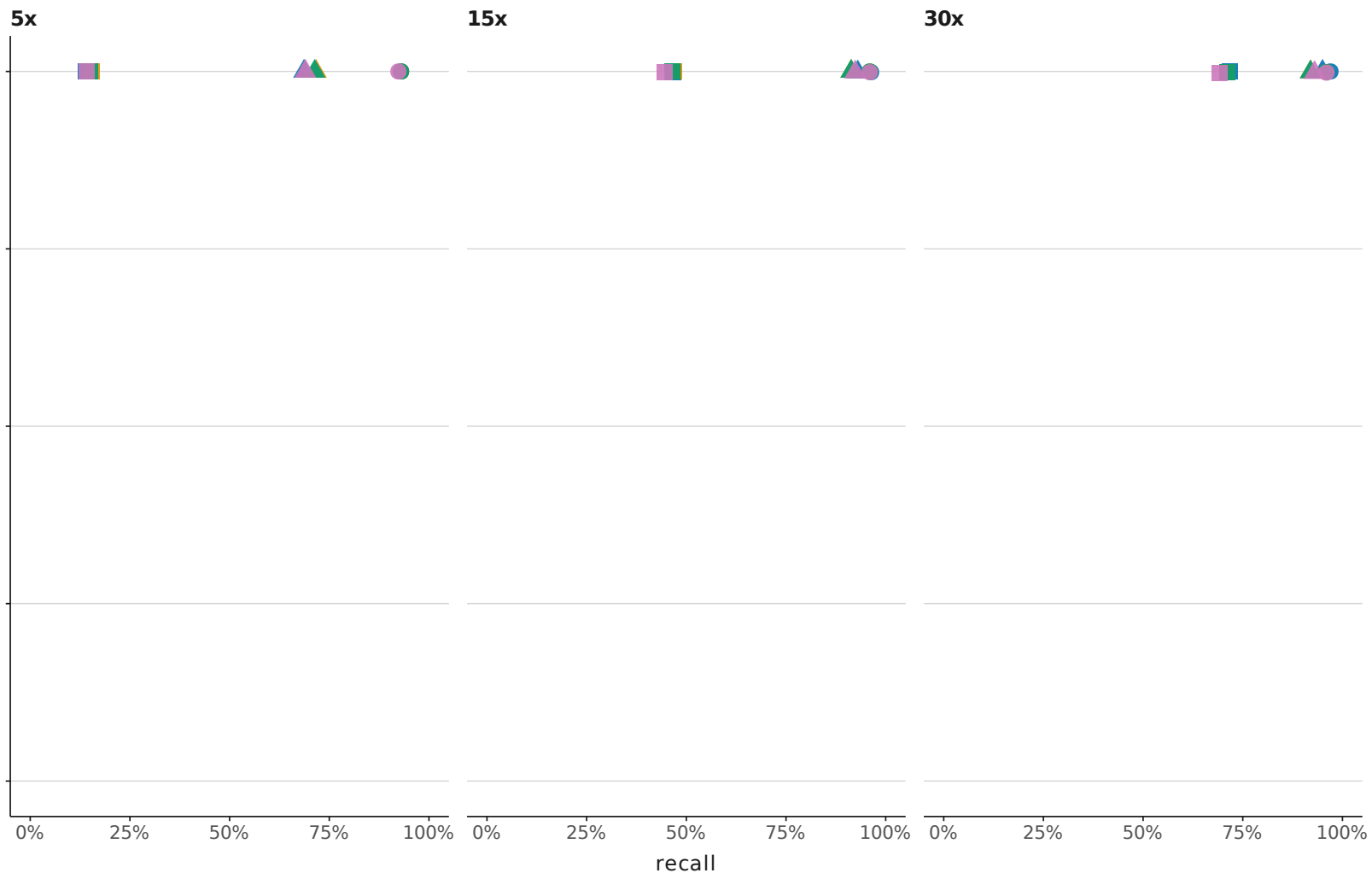
RelocaTE2 RelocaTE3-bowtie2/bwa RelocaTE3-bwa/bwa RelocaTE3-minimap2/minimap2



Precision-recall operating points

Precision = matched / all calls per sample (global denominator); one point per class

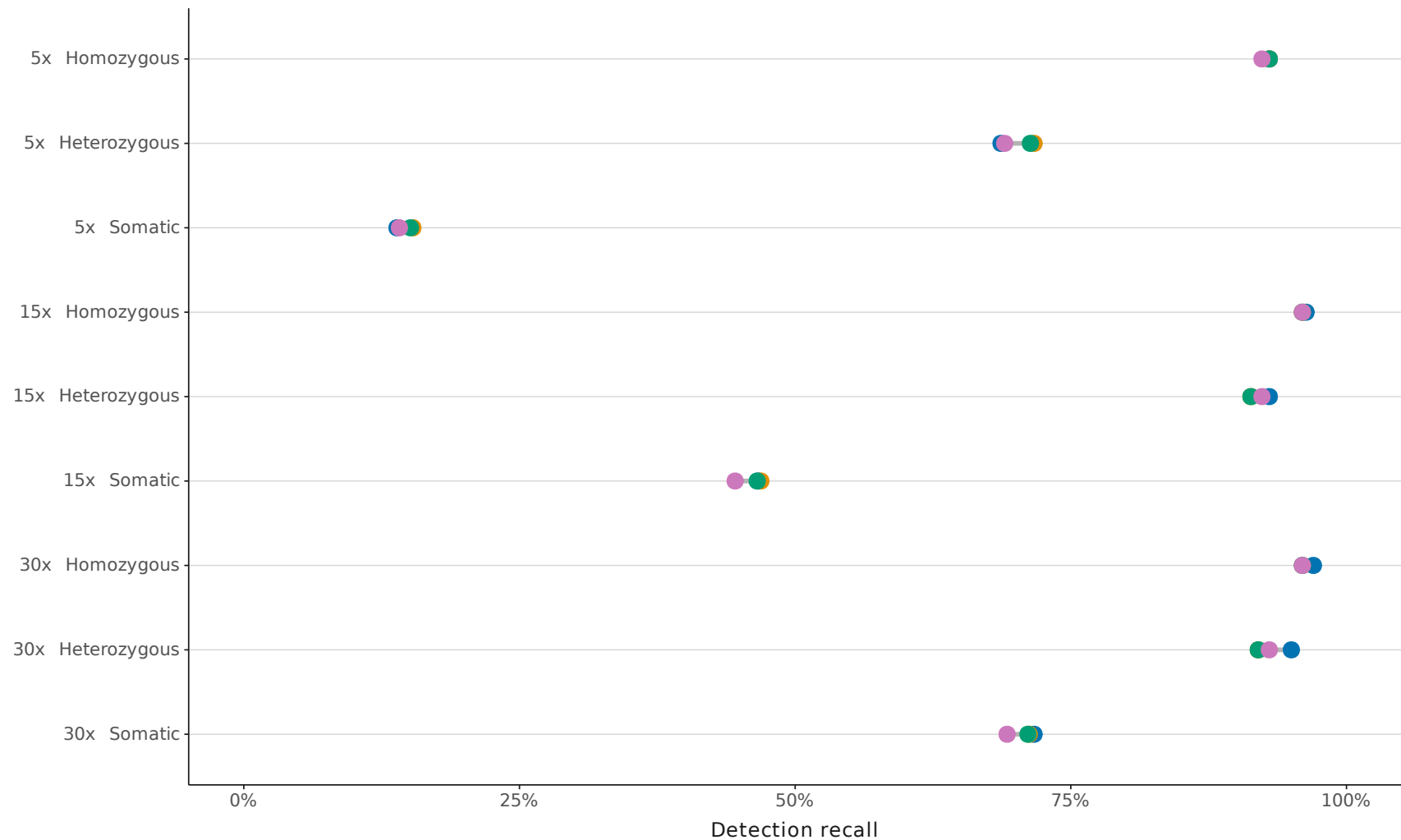
● Homozygous ▲ Heterozygous ■ Somatic ● RelocaTE2 ● RelocaTE3-bowtie2/bwa ● RelocaTE3-bwa/bwa ● RelocaTE3-minimap2/minimap2



Detection recall by caller

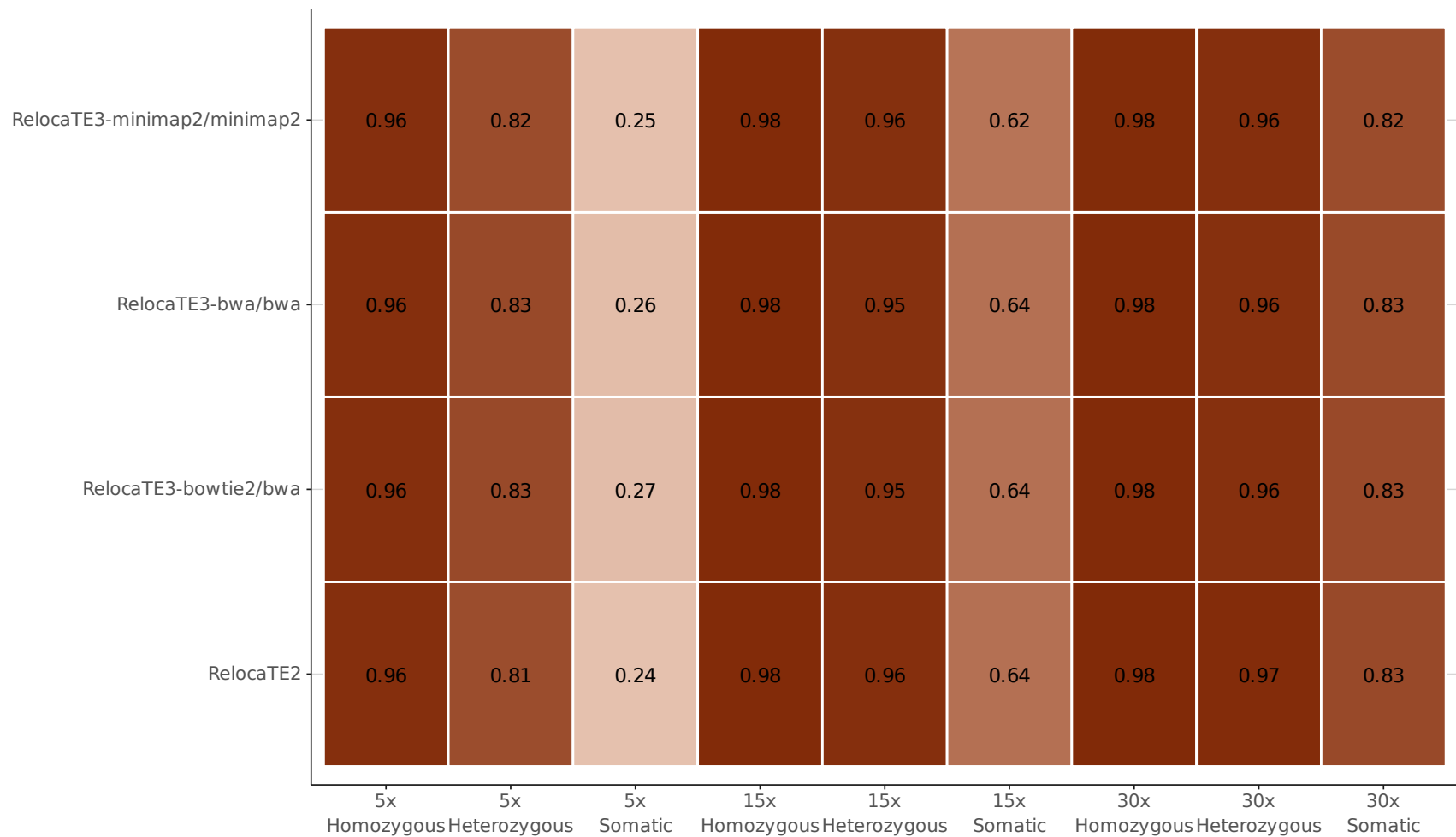
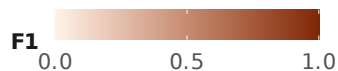
Each row = coverage x class; grey connector spans the 4 callers

● RelocaTE2 ● RelocaTE3-bowtie2/bwa ● RelocaTE3-bwa/bwa ● RelocaTE3-minimap2/minimap2



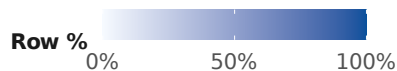
F1 (detection x precision) by condition

Harmonic mean of recall and precision (precision = global-denominator, per sample)



Genotype / detection confusion

Row-normalised: where each truth class's events land (pooled over coverage)



RelocaTE2

Truth class	Het	Hom	Hom-exc	Somatic	Missed
Somatic	917 34%	23 1%	1 0%	249 9%	1510 56%
Heterozygous	544 60%	64 7%	162 18%	0 0%	130 14%
Homozygous	2 0%	849 94%	8 1%	0 0%	41 5%

RelocaTE3-bowtie2/bwa

Het	Hom	Hom-exc	Somatic	Missed
857 32%	13 0%	1 0%	330 12%	1499 56%
610 68%	69 8%	85 9%	1 0%	135 15%
0 0%	853 95%	2 0%	0 0%	45 5%

RelocaTE3-bwa/bwa

Truth class	Het	Hom	Hom-exc	Somatic	Missed
Somatic	851 32%	12 0%	1 0%	331 12%	1505 56%
Heterozygous	610 68%	69 8%	84 9%	1 0%	136 15%
Homozygous	0 0%	853 95%	2 0%	0 0%	45 5%

RelocaTE3-minimap2/minimap2

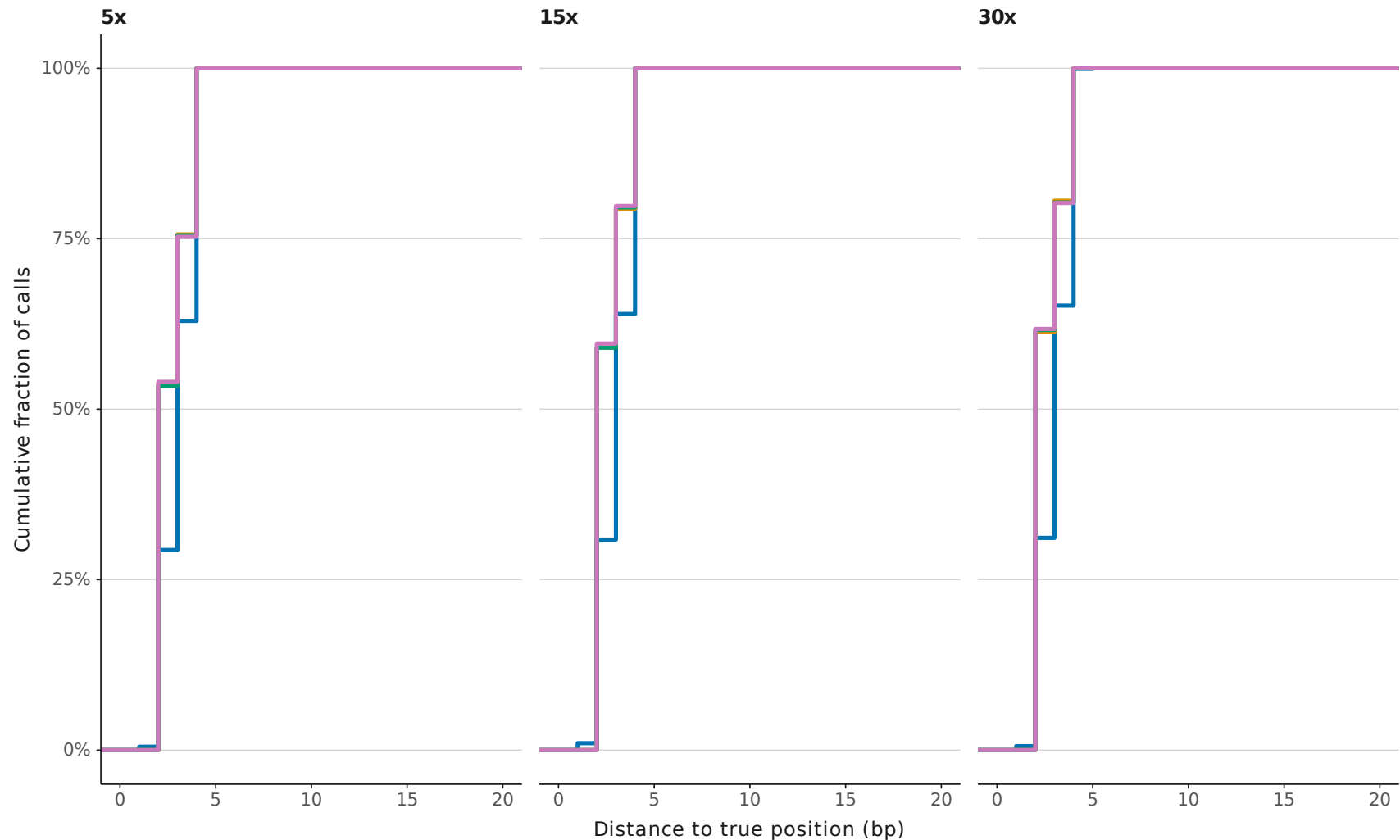
Het	Hom	Hom-exc	Somatic	Missed
805 30%	11 0%	0 0%	335 12%	1549 57%
620 69%	68 8%	74 8%	1 0%	137 15%
0 0%	850 94%	3 0%	0 0%	47 5%

Called status

Breakpoint positional accuracy

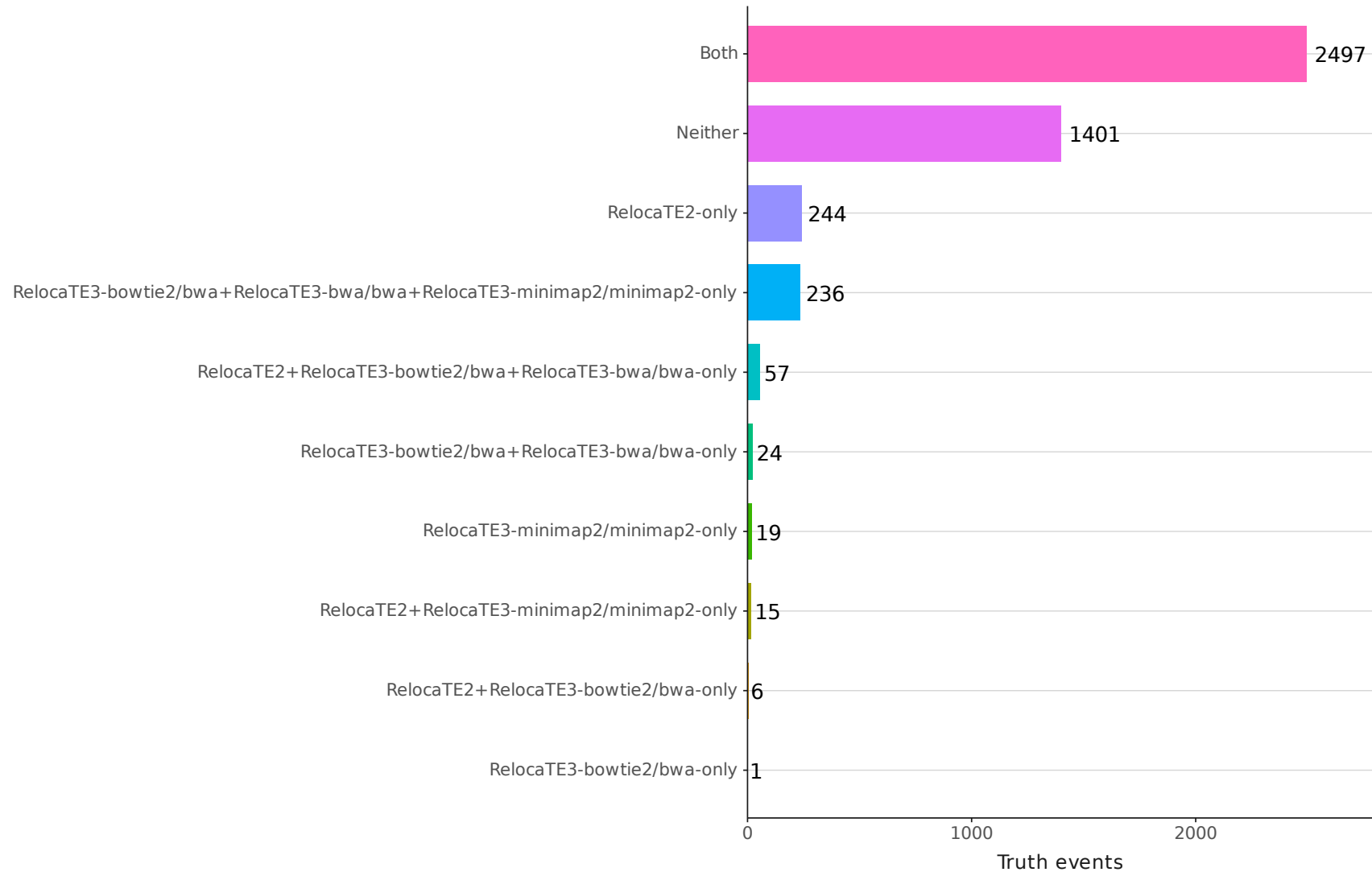
ECDF of $|\text{called} - \text{true}|$ position for matched events; steeper/left = better

RelocaTE2 RelocaTE3-bowtie2/bwa RelocaTE3-bwa/bwa RelocaTE3-minimap2/minimap2



Per-event caller agreement

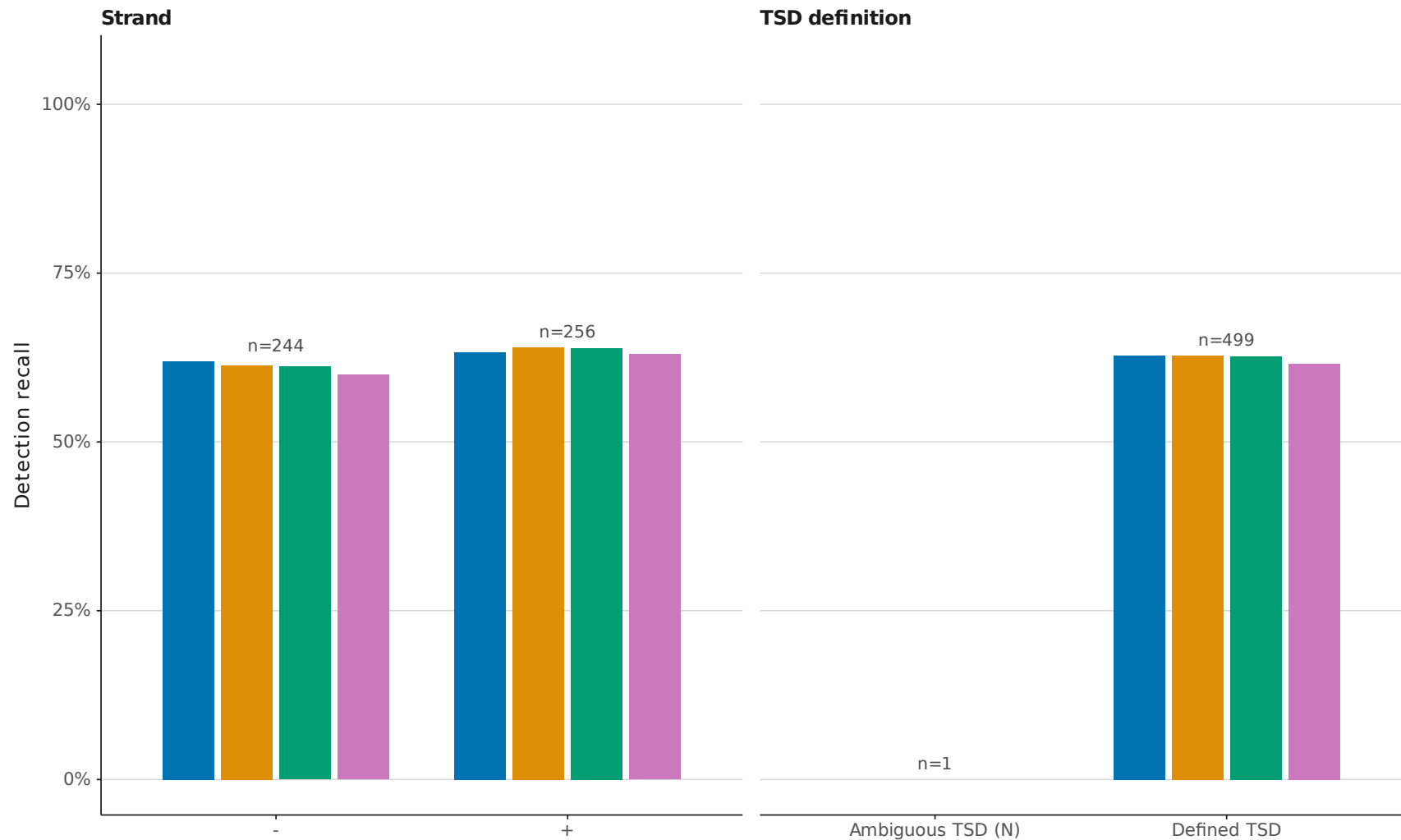
Truth events by which caller(s) detected them (pooled over all samples)



What do missed events have in common?

Detection recall stratified by strand and TSD ambiguity (pooled); n = distinct truth events

RelocaTE2 RelocaTE3-bowtie2/bwa RelocaTE3-bwa/bwa RelocaTE3-minimap2/minimap2



Compute resources vs. coverage

Per-sample wall time and peak RSS (mean $\pm$ SEM)

